# HARKIRAT SINGH

+91-8847654698 — hkkirat25@gmail.com — LinkedIn — GitHub

## Education

**Thapar Institute of Engineering and Technology**, Patiala — Patiala, Punjab
*Bachelor of Engineering in Computer Engineering* — 2023 – 2027

## Work Experience

**Summer Project Intern —Experiential Learning Centre, TIET** — June 2025 – July 2025

- Developed a passive, continuous authentication system for mobile banking using behavioral biometrics
- Implemented a Split Federated Learning architecture processing activations locally to keep raw sensor data on-device.
- Processed multi-modal mobile sensor streams including Accelerometer, Gyroscope logs, Keystroke dynamics, and Touch heatmaps to analyze user interaction patterns.

## Skills

**Programming Languages:** Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3
**Core Computer Science:** Data Structures & Algorithms, OOPS, DBMS, OS, Computer Networks
**Software Engineering:** REST APIs, API Integration, Authentication (JWT, OAuth), WebSockets
**Backend & Web Technologies:** React.js, Node.js, Express.js, FastAPI
**Databases:** PostgreSQL, MongoDB, MySQL, Redis
**Generative AI:** Scikit-learn, NLP, Langchain, Prompt Engineering, Vector Databases
**Developer Tools:** Git, GitHub, GitHub Actions (CI/CD), Docker (Basics), Linux (Basics), Vercel, Postman
**Professional Skills:** Analytical Problem Solving, Communication, Collaboration, Leadership, Adaptability,

## Projects

**KhataBox — *Inventory & B2C Retail Management Platform*** — Deployed Link & GITHUB
Tech Stack: Next.js, React, TypeScript, FastAPI, PostgreSQL, SQLAlchemy, scikit-learn

- Built a full-stack retail platform with 20+ modules, 30+ REST APIs, and JWT-based role authentication.
- Developed Random Forest-based demand forecasting to automate weekly inventory restocking using sales history.
- Implemented real-time inventory tracking with QR/barcode scanning, low-stock alerts, and automated PDF invoicing
- Engineered a scalable FastAPI + PostgreSQL backend with Redis caching and responsive Next.js dashboards.

**Symptom Score AI — *ML-Based Symptom Intelligence System*** — LINK
Next.js, TypeScript, Python, FastAPI, MongoDB, Clerk Scikit-learn, LangChain, ChromaDB

- Built an AI symptom checker with real-time disease prediction, severity analysis, and personalized recommendations.
- Developed a 3-model ML ensemble with SHAP explainability, achieving ˜89% prediction accuracy.
- Integrated a LangChain-based RAG assistant over 12 medical documents for context-aware AI guidance.
- Implemented authentication, dashboards, prediction history, and PDF/CSV reporting through 31 REST APIs.

**IntelliMoney — *AI-Powered Financial Health Scoring Platform*** — LINK
Python, React.js, FastAPI, MongoDB, Scikit-learn, NLP

- Built an AI-powered financial analytics platform with expense tracking, budgeting,& personalized financial health scoring.
- Developed NLP-based expense categorization using TF-IDF + Logistic Regression and integrated secure REST APIs, JWT authentication, and MongoDB for scalable data management.
- Designed interactive dashboards with real-time spending insights and financial recommendations.

## Achievements

- Earned a certificate in Web Development, Software Engineering, Robotics — Certifications
- Solved 250+ Data Structures and Algorithms problems on platforms like LeetCode, GeeksforGeeks.
- Organized and executed events for 450+ participants, coordinating logistics and leading a team of 27 members.

## Positions of Responsibility

**Joint Secretary– IETE Students Forum, TIET Patiala** — May 2025 – May 2026

- Coordinated 10+ tech events and workshops, increasing club membership by 40%

**Student Ambassador— LetsUpgrade EdTech Pvt. Ltd.** — August 2024

- Onboarded 45+ students into skill-building workshops via targeted promotion.
- Led social outreach initiatives, boosting referral engagement by 20%.

**Creative Content Head— Vyom Hostel TIET (Semester-6)** — January 2026 - May 2026